Target Paper Ranking: 3

Error Analysis:

(1) **Formatting.** As outlined in Section 3.3, a notable limitation of open-sourced LLMs is their ability to format desired output. During the third round of Qwen response, though the ground truth paper is ranked 3rd in the additional texts, it is not included in the formatted Python dictionary as requested by the prompt. This incident highlights its constrained formatting capabilities. As a corrective measure, each Qwen response is subsequently processed through GPT-3.5/GPT-4 for further formatting before the next round of ranking.

(2) **Text Analysis.** Due to Qwen's limited token length, reference papers are divided into batches of 8 for ranking. Each batch requires the model to restate the visual context question in its formatted output, which should remain consistent across batches. However, inconsistencies are observed as the question differs in Round 2 Batch 00. Despite Qwen's high performance across several existing benchmarks, it is hypothesized that the scientific figure input has compromised its text analysis capabilities, resulting in hallucinatory results.

(3) **Figure Analysis.** Variations in the responses to the visual context question across different rounds and batches suggest that Qwen's scientific figure analysis capabilities are unstable and may be influenced by textual inputs.

(4) **Prompt Limitation.** The wording used to direct the model to re-rank papers based on their "relevance" to the question may introduce ambiguity in the inference process. This can lead the model to prioritize the topic of the question over the accuracy of the answers. For example, in Round 3, the model potentially emphasized the keyword "hallucination," associated with the Hallucination Vulnerability Index (HVI) score, over "T5," affecting the ranking outcomes.

(5) **Batch Limitation.** The division of reference papers into batches of 8 could result in varying batch size in the last batch, which ranges from 1 to 8 papers. It is conjectured that smaller batches may provide an advantage in ranking accuracy. In this case, with 41 reference papers, the last batch contains only one paper. Although the model incorrectly answered "T5" for the visual context question, it still ranked this single paper, leading to a suboptimal outcome where the ground truth paper was ranked third. This factor potentially contributes to the outcome where the ground truth paper was ranked 3rd despite the incorrect answer to the question.

Category: Wrong answer, Suboptimal ranking